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Department of Electrical And Electronic Engineering

Lab report 1

Course Code: EEE-1214

Course Title : Circuit 2 Lab

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Class Task:

1. **Draw the RL series circuit and observe the step response of the constructed circuit. Analyze the waveshapes of the voltage and current and archive the screenshots of the outputs.**

Resistor = 1000 Ohms

Inductor = 200 Henry

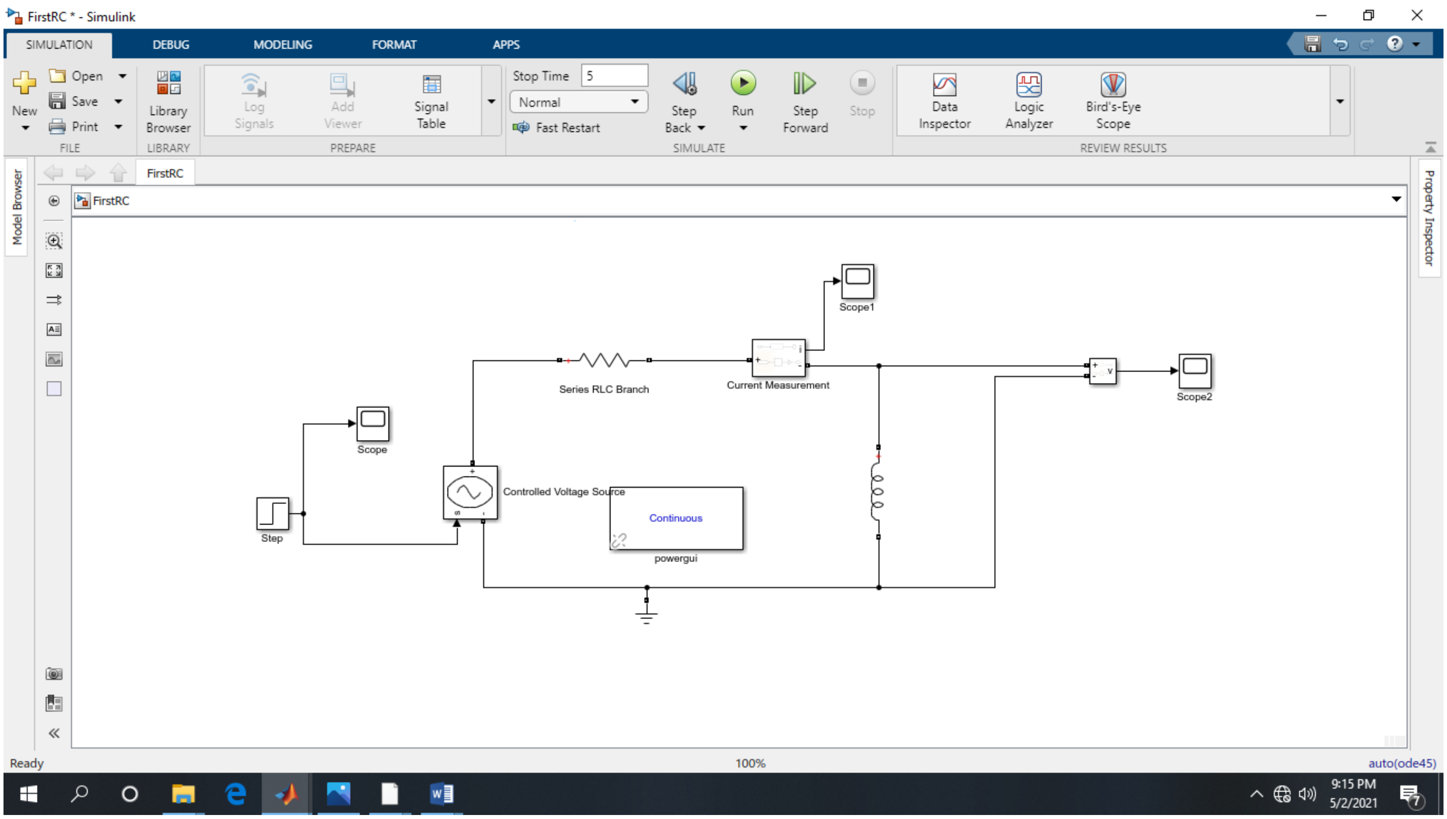

Fig 1: Circuit Diagram (RL Circuit)

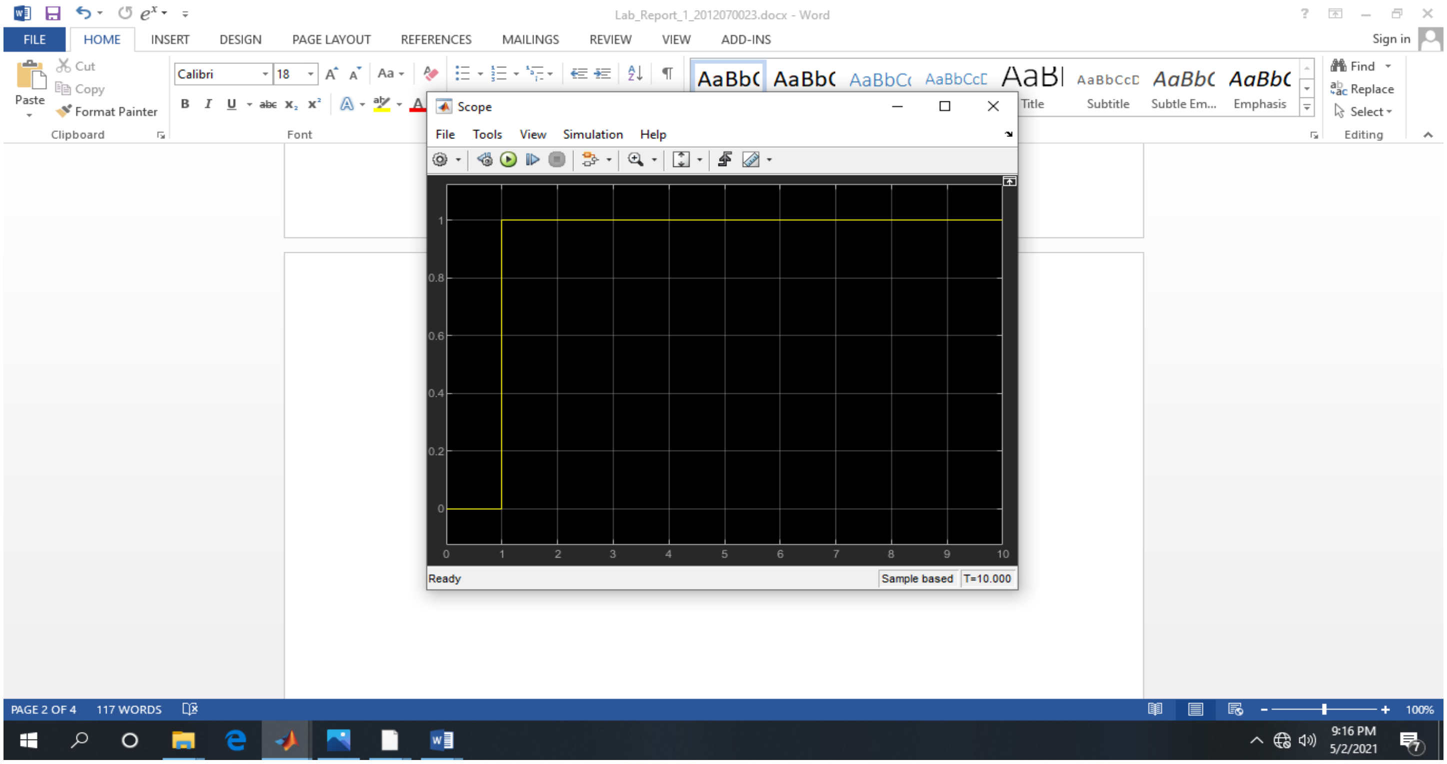

Fig 2: Scope (Wave shape of step Function)

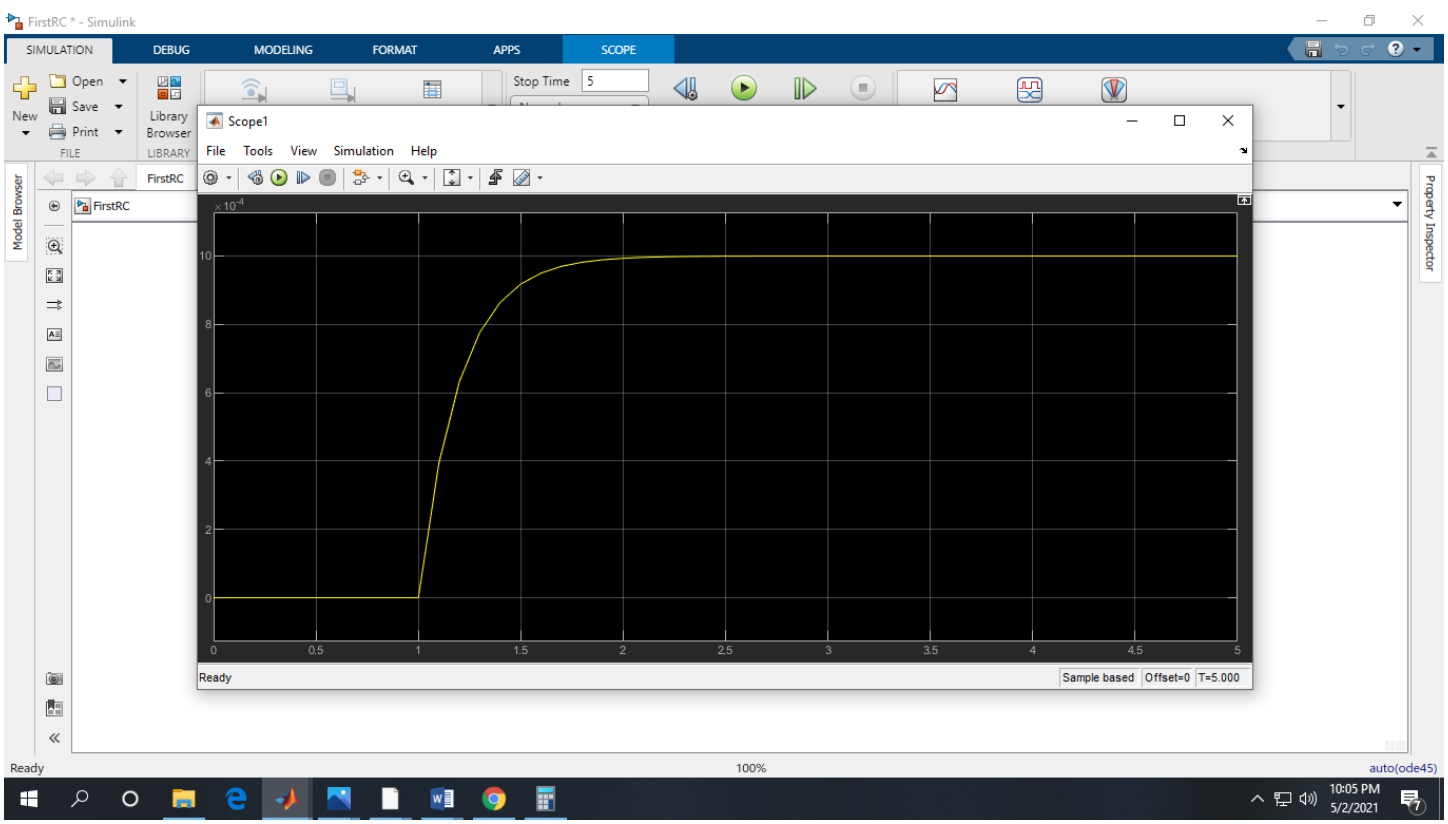

Fig 3: Scope 1(Wave shape of Current)

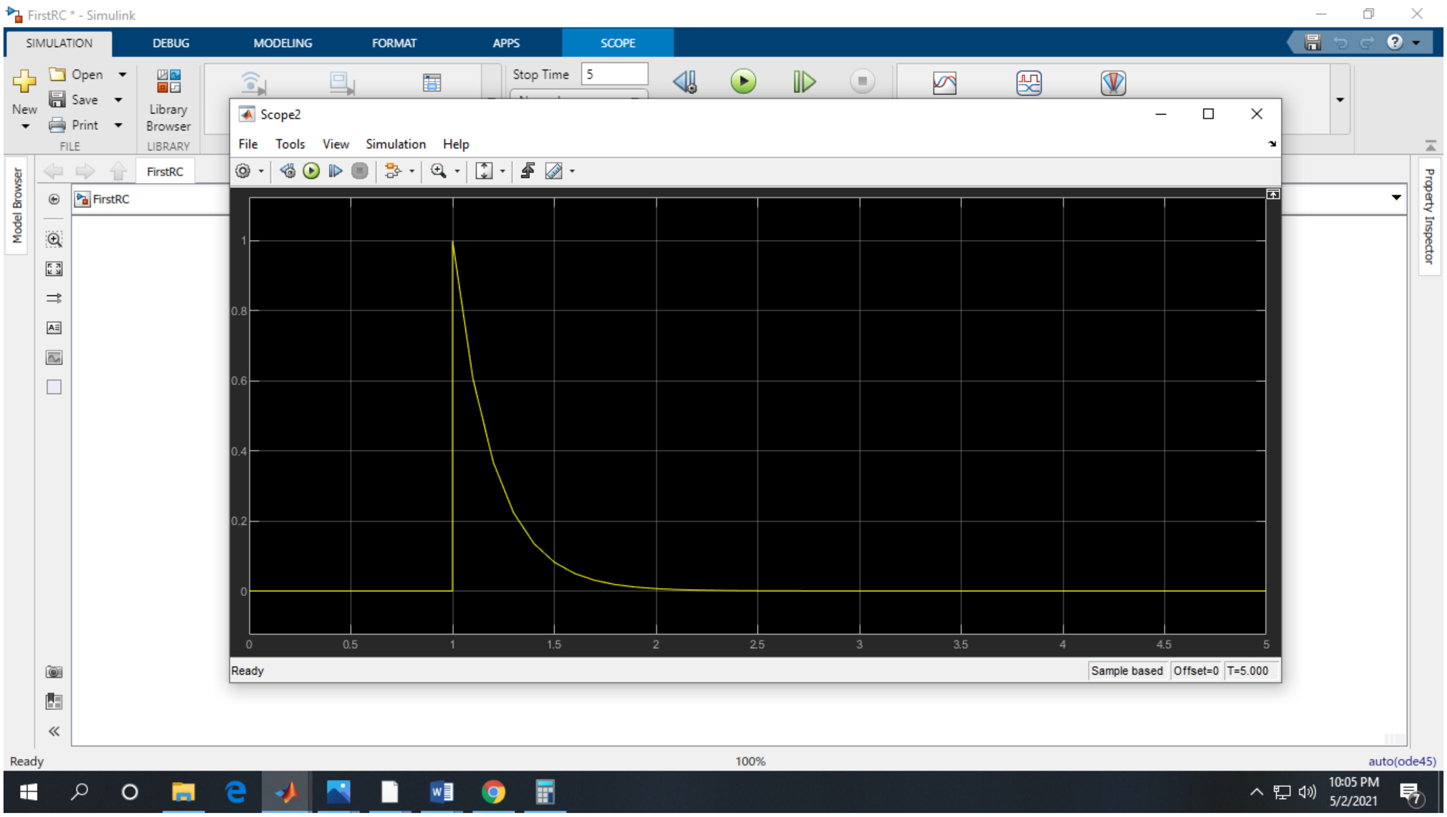

Fig 4: Scope 2(Wave shape of Voltage across Inductor)

Discussion: A circuit that contains a pure resistance R ohms connected in series with a coil having a pure inductance of L (Henry) is known as RL Series Circuit**.** Here we use A Voltage source (DC), a Resistor, a Inductor and a Step Function Our goal is to find inductor current i as the circuit response. Rather than apply KCL,

$$\mathbf{i = i_t + i_{ss}}$$

Here, τ it is a time constant. The RL time constant indicates the amount of time that it takes to conduct 63.2% of the current that results from a voltage applied across an inductor. In other words, the RL time constant in seconds is equal to the inductance in henrys divided by the resistance of the circuit in ohms. So in our RL circuit the time constant is $\tau = L/R = 200/1000 = 0.2$s. that means, In 0.2 second to conduct 63.2% of the current that results from a voltage applied across an inductor.

Before 1 second the step function is 0 and its transition at 1 second. After one second the step function will increase and after infinite time the inductor act like short circuit so after infinite time the voltage (V_L) across the inductor is zero.

i(t=1s) = 0	V_L(t=1s)=V
i(t=infinite)=V/R	V_L(t=infinite)=0

In Fig 3: Scope 1(Wave shape of Current) here Y axis is i(t)=V/R and the X axis is time (t).

$i(t) = V/R\ (1-e^{-(t/\tau)})$

$= 10 \times 10^{-4}(1-e^{-t/.2})$

Here , i= V/R = 10×10^{-4}A we get this value from Graph (Fig:3).then if we assign the value in t then we get a graph line like fig:3. And it's the shape of current increasing when input voltage.

<u>Class task 2:</u>

1. **Following the exact same steps as Part C, simulate a Series RLC circuit and observe the different voltage and current wave shapes.**
2. **Save the wave shapes as screenshots and show them to your teacher.**

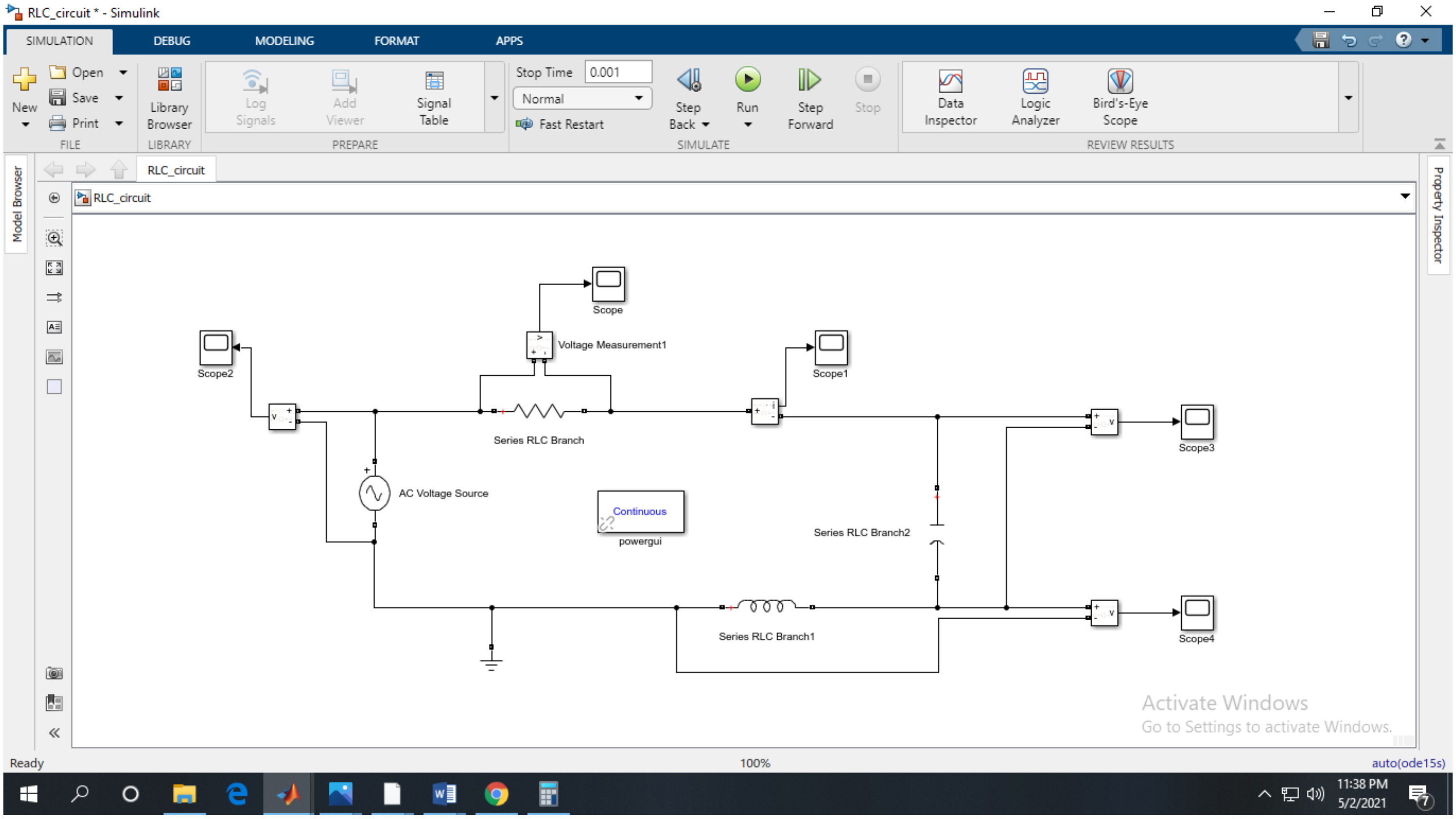

Figure 2(a): **Series RLC circuit**

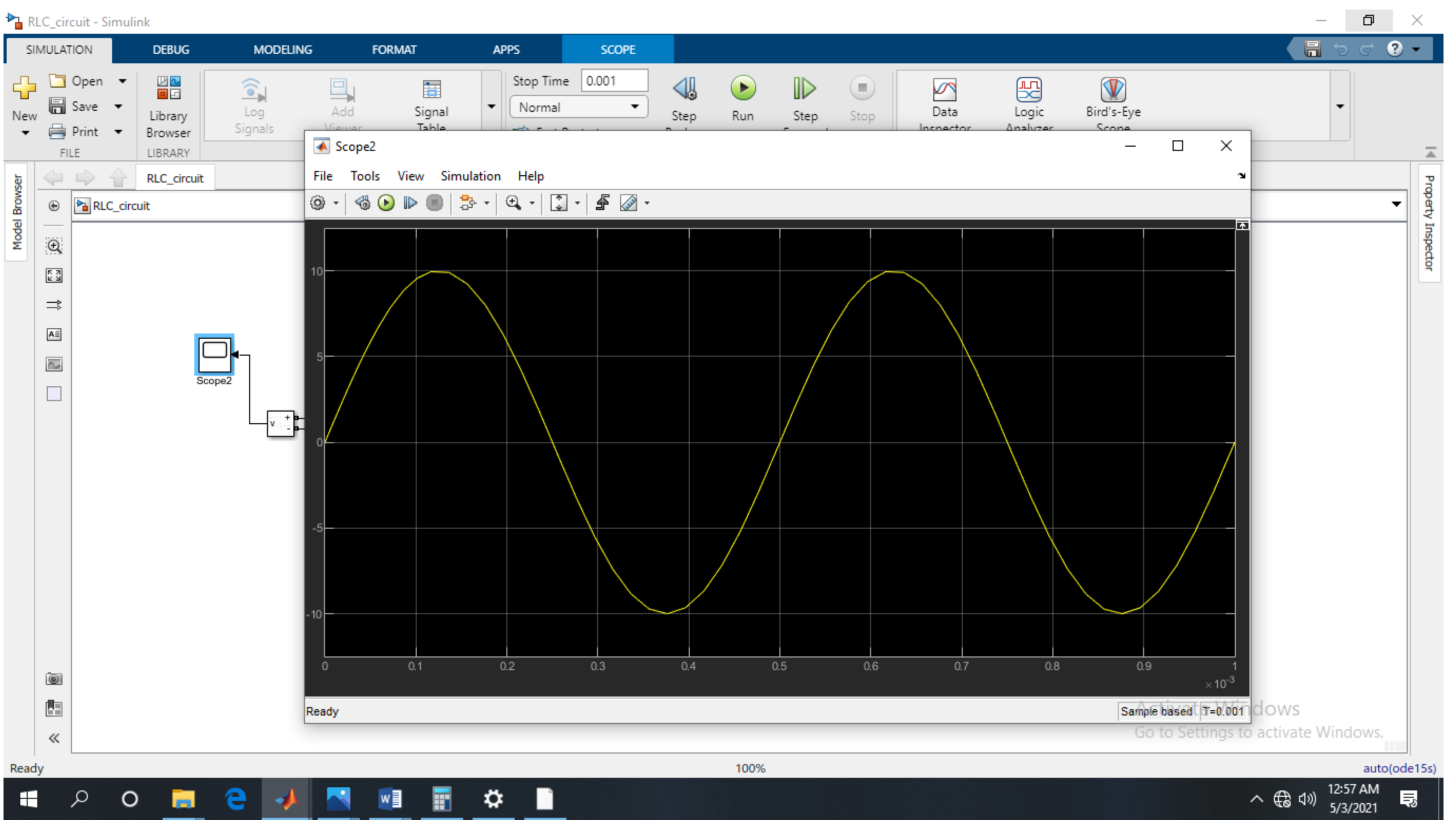

Figure 2(b): (Scope 2) Wave shape of Voltage (AC Voltage Source)

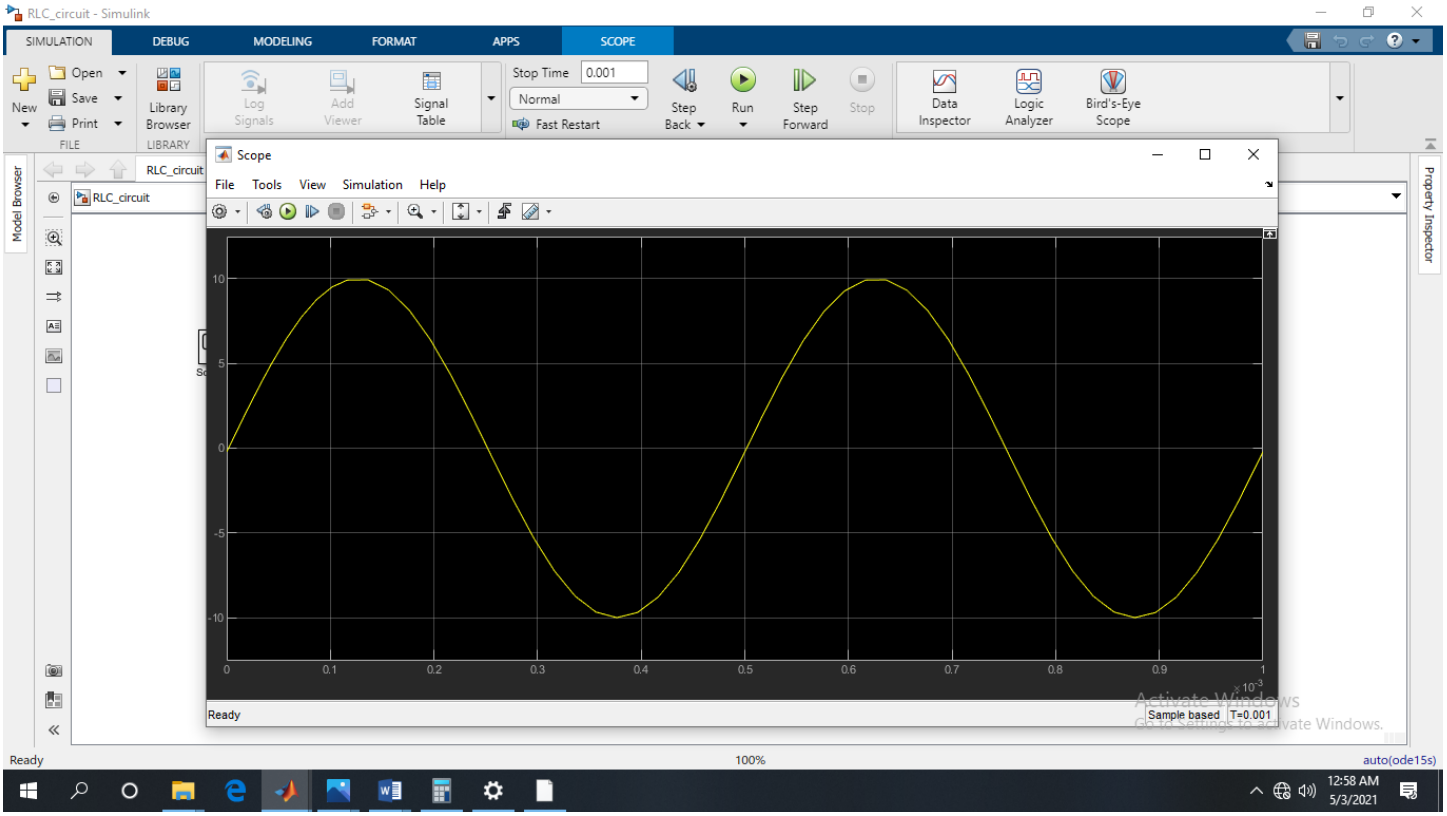

Figure 2(c): (Scope) Wave shape of Voltage across resistor

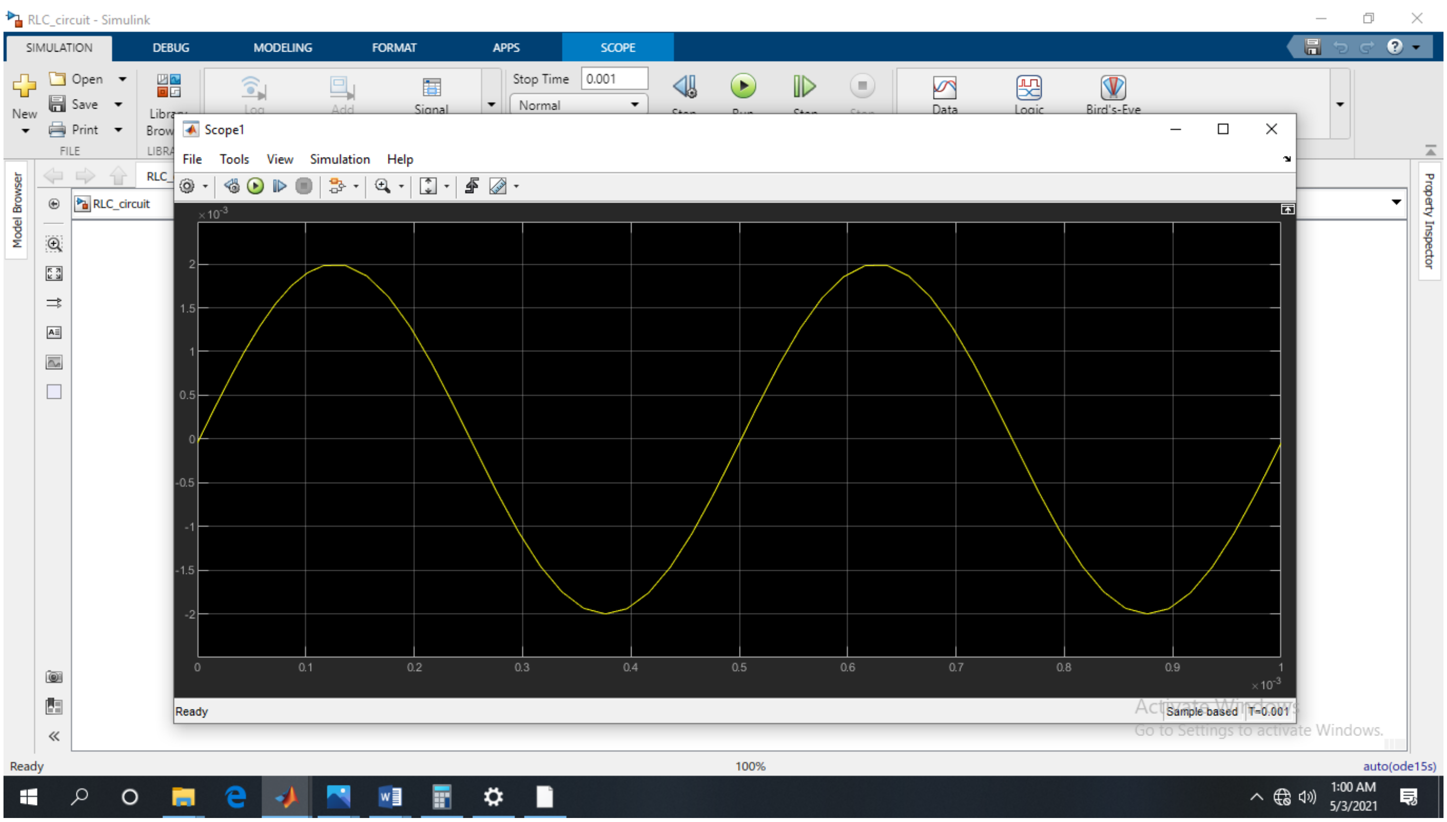

Figure 2(d): (Scope1) Wave shape of Current in series RLC circuit

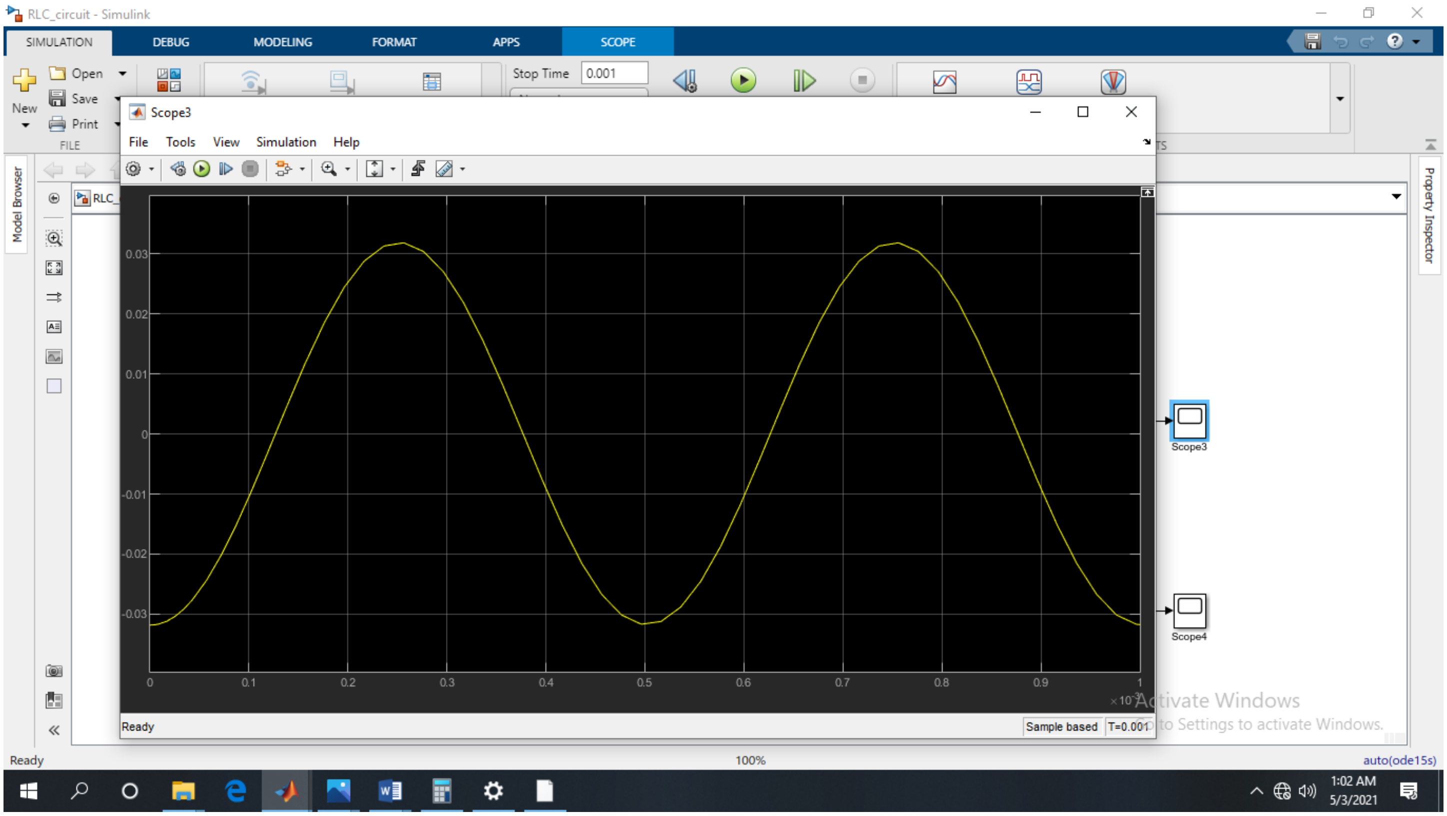

Figure 2(e): (Scope3) Wave shape of voltage across Capacitor

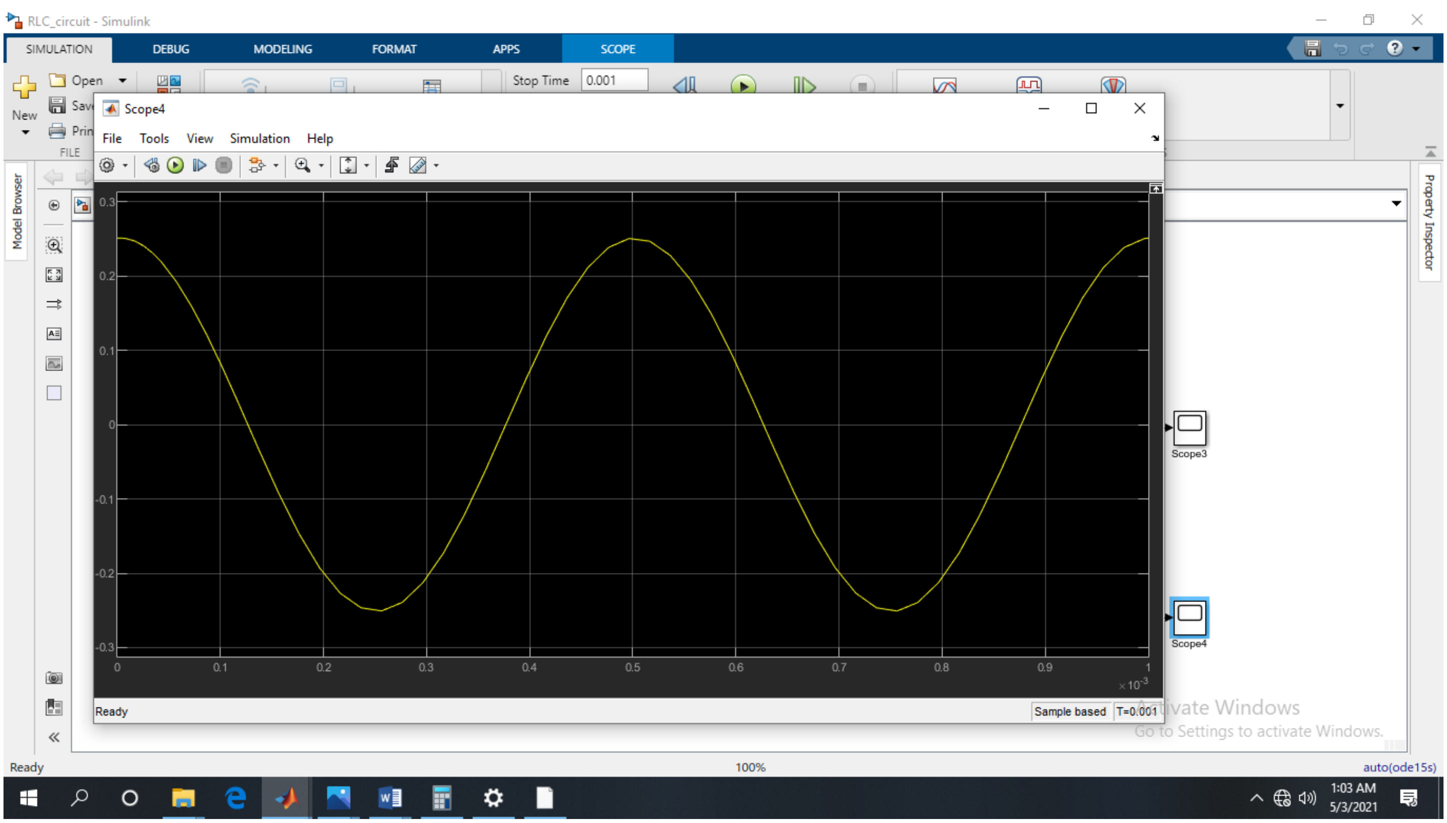

Figure 2(f): (Scope4) Wave shape of voltage across Inductor

Discussion:

We observe from wave shapes of current and voltage across resistor, capacitor and inductor' the peak value and peak to peak value.

Here In Figure 2(b): (Scope 2) Wave shape of Voltage (AC Voltage Source) we observe voltage wave shape and find its peak value and peak to peak value.

Voltage Source:

The peak value or amplitude of wave shape of this voltage source is 9.941V

The peak to peak value of wave shapes of this voltage source is 19.46V

Similarly,

In Figure 2(c): (Scope) Wave shape of Voltage across resistor

Hear, the voltage drop on the resistor. From this wave shape (fig(2c)) we get peak value and peak to peak value of voltage.

Resistor:

The peak value or amplitude of wave shape of this resistor is 9.926V

The peak to peak value of wave shapes of this resistor is 19.92V

Similarly,

In Figure 2(d): (Scope1) Wave shape of Current in series RLC circuit

From this wave shape of current, we get peak value and peak to peak value.

Current:

The peak value or amplitude of wave shape of the current is 1.985×10^{-3} A

The peak to peak value of wave shapes of the current is $3.985x10^{-3}$ A

In Figure 2(e): (Scope3) Wave shape of voltage across Capacitor

From this wave shape of voltage across capacitor, we get peak and peak to peak value of this voltage wave shape.

Capacitor:

The peak value or amplitude of wave shapes of the voltage across capacitor is $3.184x10^{-2}$ V

The peak to peak value of wave shapes of the voltage across capacitor is $6.366x10^{-2}$ V

In Figure 2(f): (Scope4) Wave shape of voltage across Inductor

From this wave shape of voltage across inductor, we get peak and peak to peak value of this voltage wave shape.

Inductor:

The peak value or amplitude of wave shapes of the voltage across inductor is $2.512x10^{-1}$ V

The peak to peak value of wave shapes of the voltage across inductor is $5.021x10^{-1}$ V